CISCO PACKET TRACER

NETWORK SIMULATION LAB DOCUMENTATION

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Tools used: Cisco Packet Tracer

Total labs: Nine 9

Purpose: Personal learning & skill development

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Executive Summery

This document presents the technical documentation of 6 structured networking labs completed using Cisco Packet Tracer. These labs were undertaken as a foundational step in building the networking knowledge required for a career in offensive security and penetration testing. The labs span core networking concepts including LAN design, VLAN segmentation, dynamic routing protocols, and access control — skills that directly underpin network-based penetration testing engagements.

Objectives

- Develop a thorough understanding of core networking concepts applicable to penetration testing
- Practice designing, configuring, and troubleshooting network topologies in a simulated environment
- Gain hands-on experience with Cisco IOS commands and device configuration
- Understand VLAN segmentation and inter-VLAN routing as a precursor to VLAN-based attack techniques
- Learn routing protocol behaviour (OSPF & RIP) to support network enumeration skills
- Implement ACLs and firewall rules to understand traffic filtering — both defensively and offensively

Scope

- All activities were performed exclusively within Cisco Packet Tracer — a network simulation software
- No physical hardware, real routers, switches, or live networks were involved
- Labs cover: Basic LAN Setup, VLAN Configuration, Routing Protocols (OSPF & RIP), and ACL/Firewall Rules
- Total of 6 individual labs completed across four core modules

Environment and Tools

Simulation Software	Cisco Packet Tracer (latest available version)
Devices Simulated	Cisco Routers, Cisco Switches, PCs, Servers
Network Types	LAN, VLAN, Routed Networks, Access-Controlled Networks
Protocols Used	OSPF, RIP v2, 802.1Q (VLAN Trunking), Standard & Extended ACL
Host OS	Windows (Packet Tracer installed locally)
Environment Type	Fully simulated — no physical hardware required

Methodology

1. **Objective Definition** — Define what the lab aims to achieve before starting
2. **Topology Design** — Plan and build the network topology in Packet Tracer before configuring devices
3. **Device Configuration** — Configure routers, switches, and end devices via Cisco IOS CLI
4. **Verification & Testing** — Test connectivity and expected behaviour using ping, traceroute, and show commands
5. **Troubleshooting** — Diagnose and resolve any configuration issues independently
6. **Documentation** — Record configurations, commands used, outcomes, and observations

Lab Walkthroughs

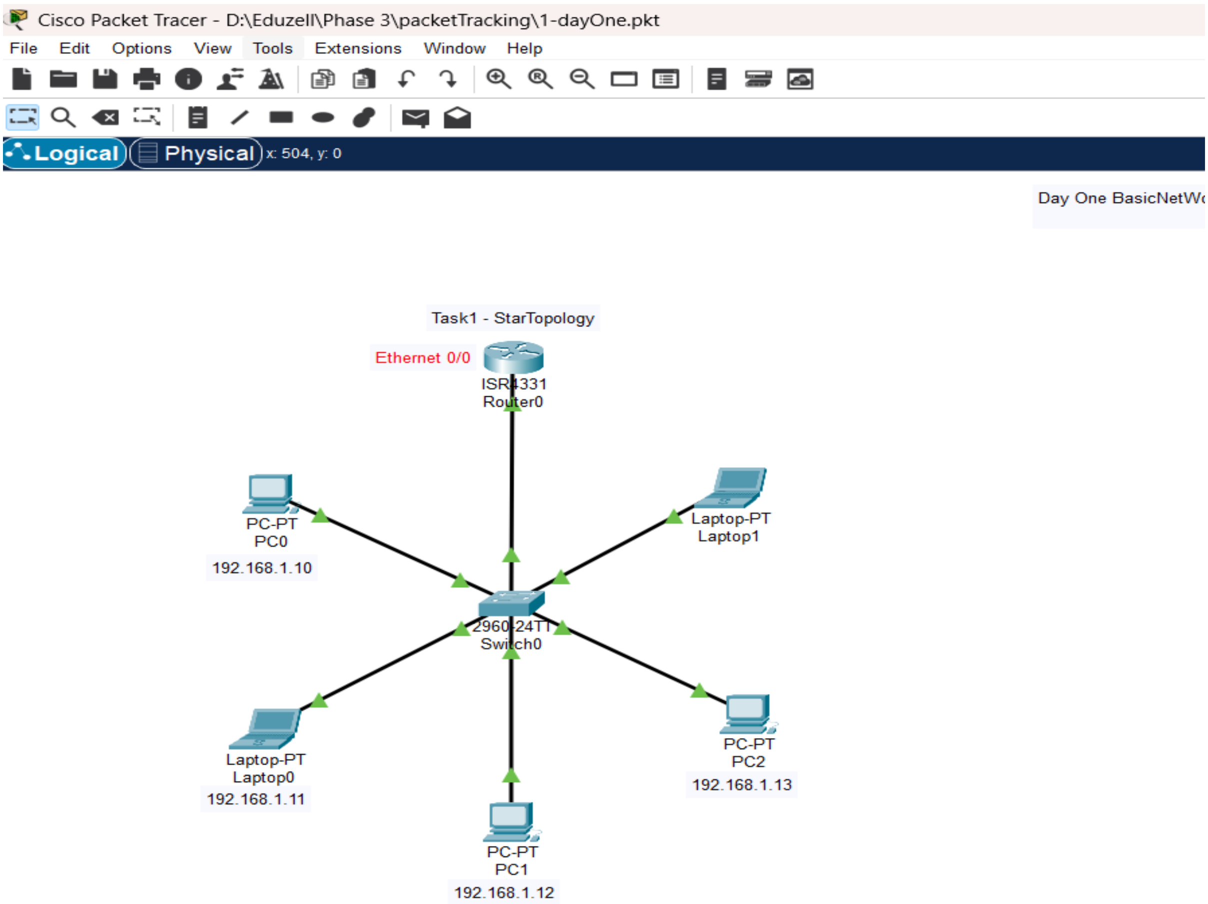
Module 1: Basic Network Setup

The foundational module covering the design and configuration of a basic Local Area Network (LAN). These labs established the core skills of IP addressing, device configuration, and connectivity verification that all subsequent labs build upon.

Lab 1.1 — Basic LAN Topology Design

Objective: Design a simple LAN topology and verify end-to-end connectivity
Devices Used: Cisco Switch, 5x PCs
Key Commands: IP address, no shutdown, ping

- Added end devices (PCs) and a switch to the Packet Tracer workspace
- Connected all devices to the switch using straight-through cables
- Assigned static IP addresses to each PC within the same subnet (e.g., 192.168.1.0/24)
- Verified connectivity between all hosts using the ping command from each PC



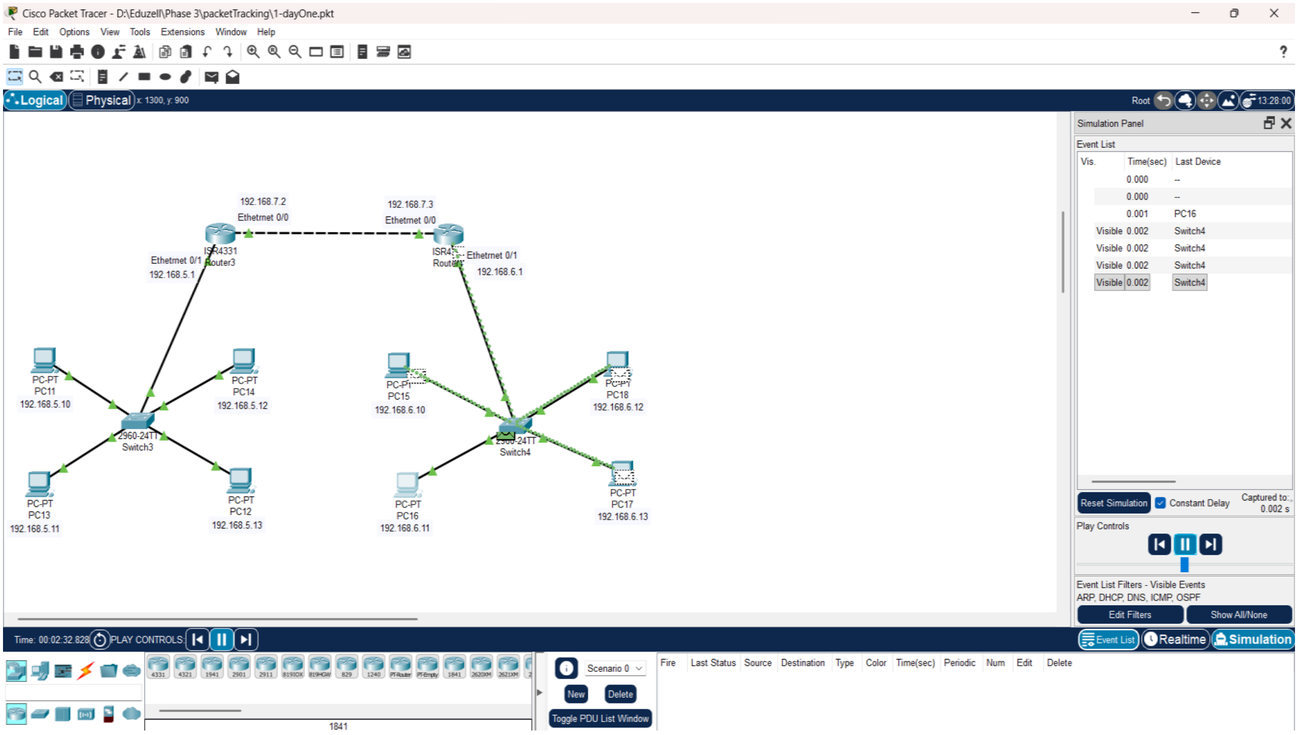
Lab 1.2 — IP Addressing & Subnetting

Objective: Practice assigning IP addresses and understanding subnet boundaries

Devices Used: Cisco Router, 2x Switches, 8x PCs

Key Commands: IP address, show IP interface brief, ping, show a

- Designed a two-subnet topology
- Assigned unique IP addresses to all hosts within their respective subnets
- Configured router interfaces with the correct gateway IP for each subnet
- Tested intra-subnet and inter-subnet connectivity to understand routing boundaries
- Used show IP interface brief to verify interface status and IP assignments



Module 2: VLAN Configuration

This module covers the segmentation of a network using Virtual Local Area Networks (VLANs). VLANs are a critical concept in both network administration and offensive security understanding VLAN architecture is essential for identifying misconfigurations such as VLAN hopping vulnerabilities.

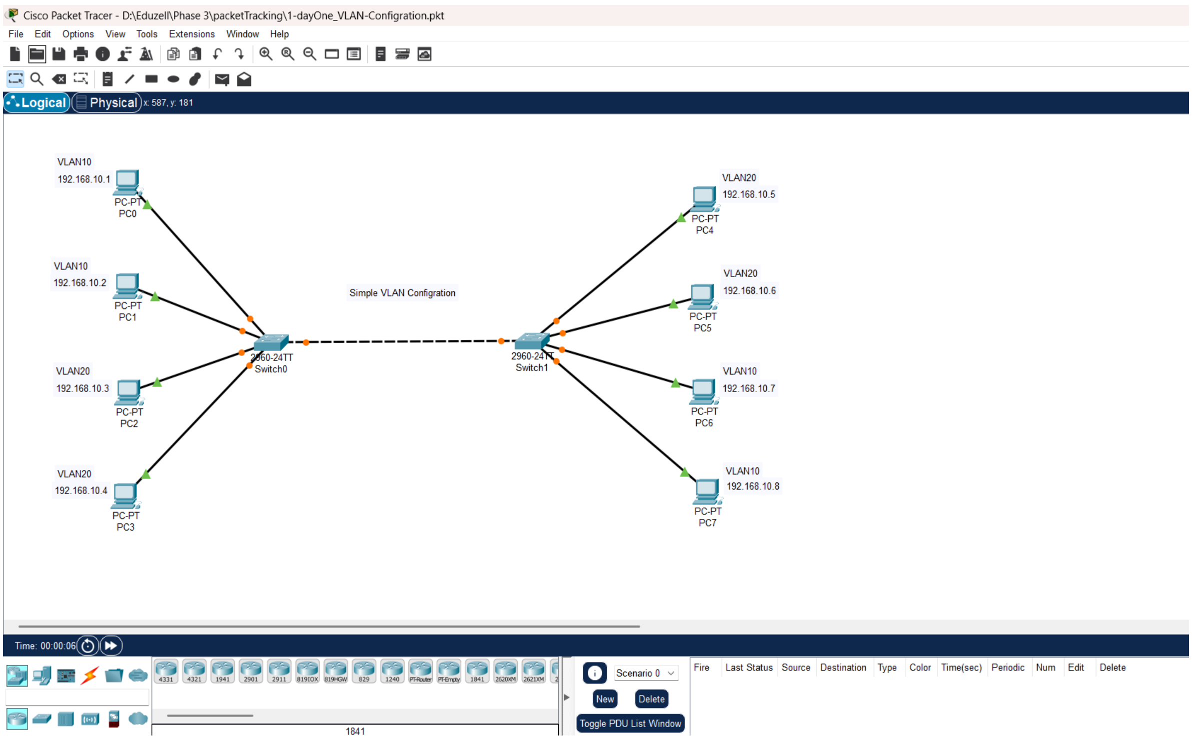
Lab 2.1 — Switch VLAN Configuration

Objective: Create and assign VLANs on a Cisco switch to segment traffic

Devices Used: 2x Cisco Switch, 8x PCs

Key Commands: name, switchport mode access, switchport access VLAN, show VLAN brief

- Created multiple VLANs (e.g., VLAN 10 — HR, VLAN 20 — IT) on a Cisco switch
- Assigned switch ports to their respective VLANs using switchport access commands
- Verified VLAN assignments using the show VLAN brief command
- Confirmed that hosts in different VLANs could not communicate without a router — validating VLAN isolation
  Tested intra-VLAN connectivity to ensure hosts within the same VLAN could reach each



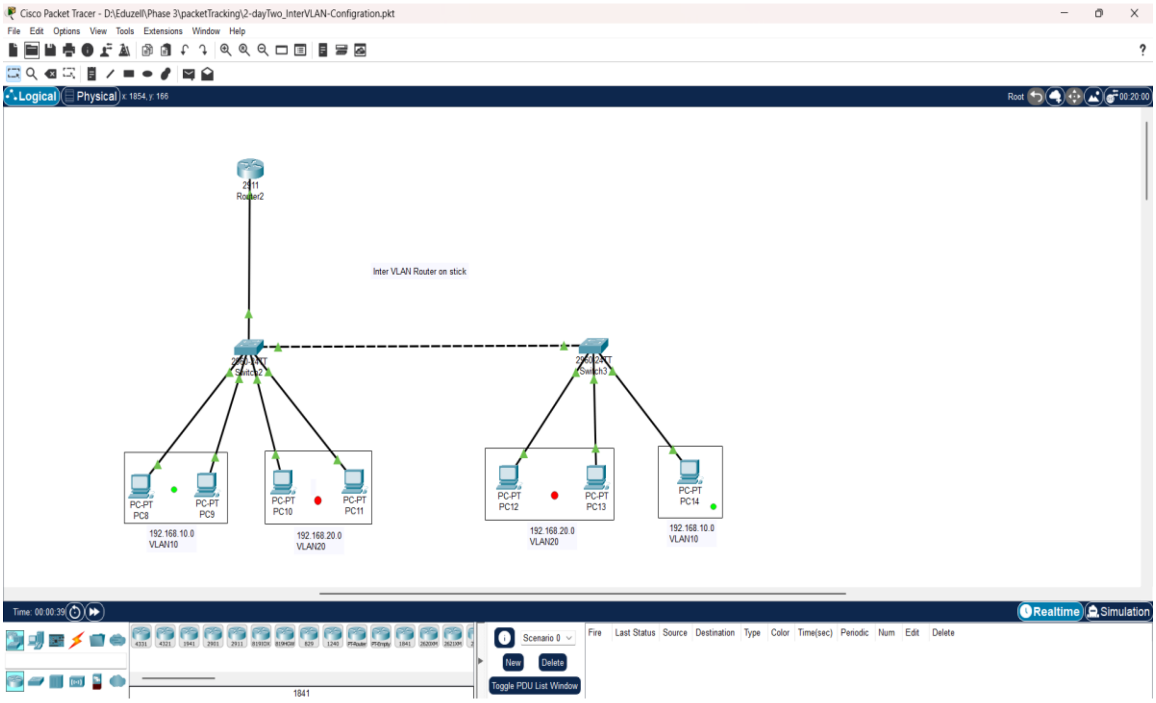
Lab 2.2 — Inter-VLAN Routing (Router-on-a-Stick)

Objective: Enable communication between VLANs using a router sub-interface configuration

Devices Used: Cisco Router, 2x Cisco Switch, 7x PCs across 2 VLANs

- Configured the switch uplink port as a trunk port to carry traffic for multiple VLANs
- Created sub-interfaces on the router for each VLAN using 802.1Q encapsulation
- Assigned a gateway IP address to each sub-interface corresponding to each VLAN subnet
- Tested inter-VLAN connectivity by pinging between hosts on VLAN 10 and VLAN 20

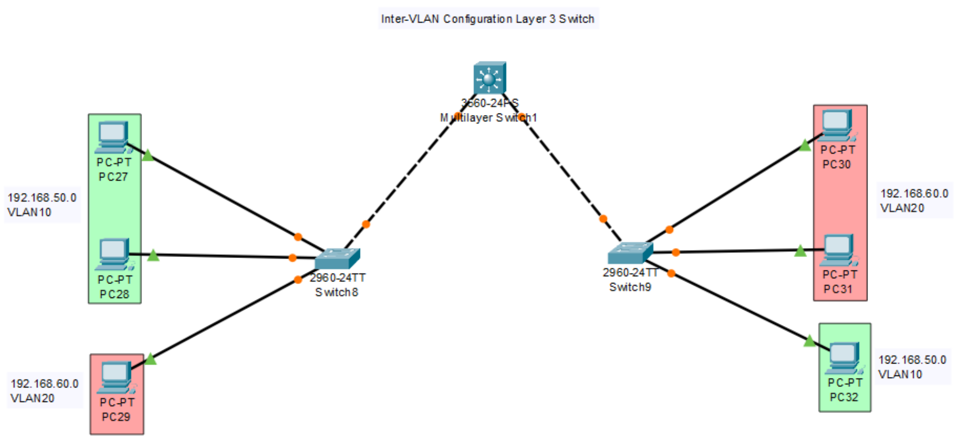
  Verified trunk port operation using show interfaces trunk on the switch



Lab 2.3 — Inter-VLAN Routing (Layer 3 Switch)
Objective: Implement inter-VLAN routing using a multilayer switch without a dedicated router
Devices Used: Cisco Layer 3 Switch, 6x PCs across 2 VLANs

- Enabled IP routing on the Layer 3 switch using the IP routing command
- Created VLAN interfaces (SVIs) and assigned gateway IP addresses to each
- Confirmed routing table entries using show IP route

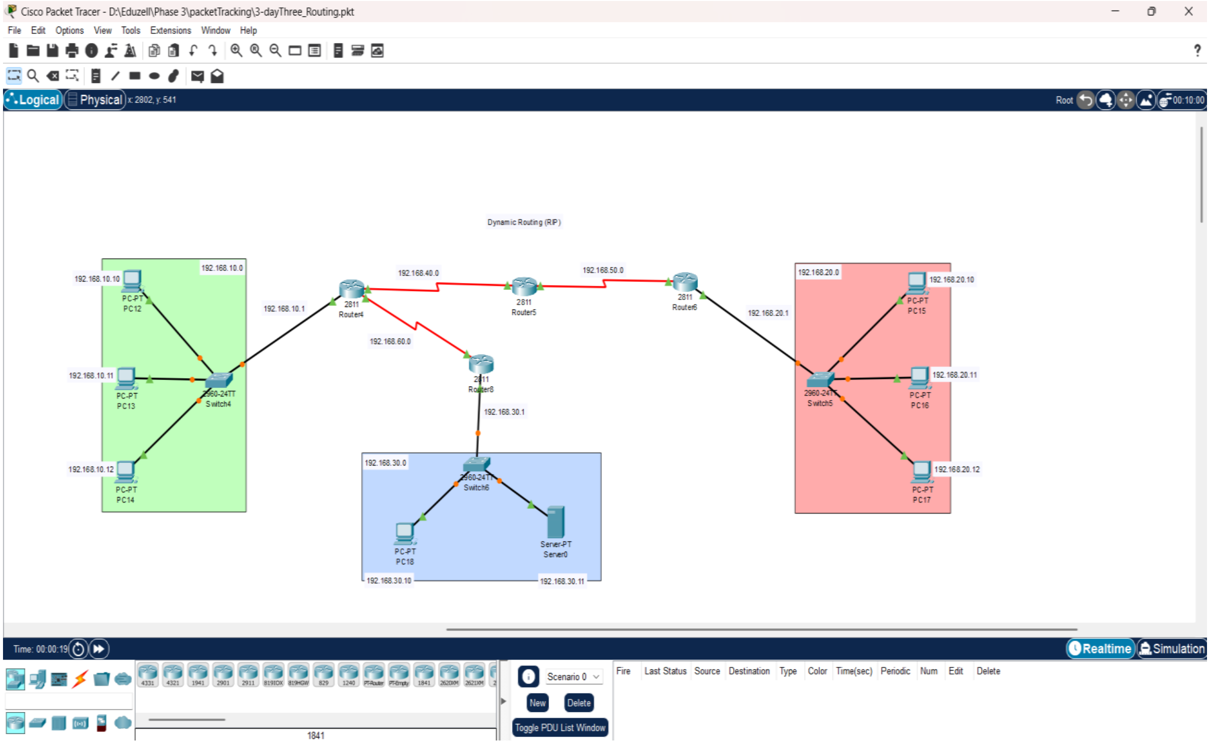
Verified end-to-end connectivity between hosts on different VLANs routed through the switch



Module 3: Routing Protocols
This module covers the configuration and analysis of dynamic routing protocols. Understanding how routers share and update routing information is fundamental to network enumeration and attack path identification in penetration testing engagements.

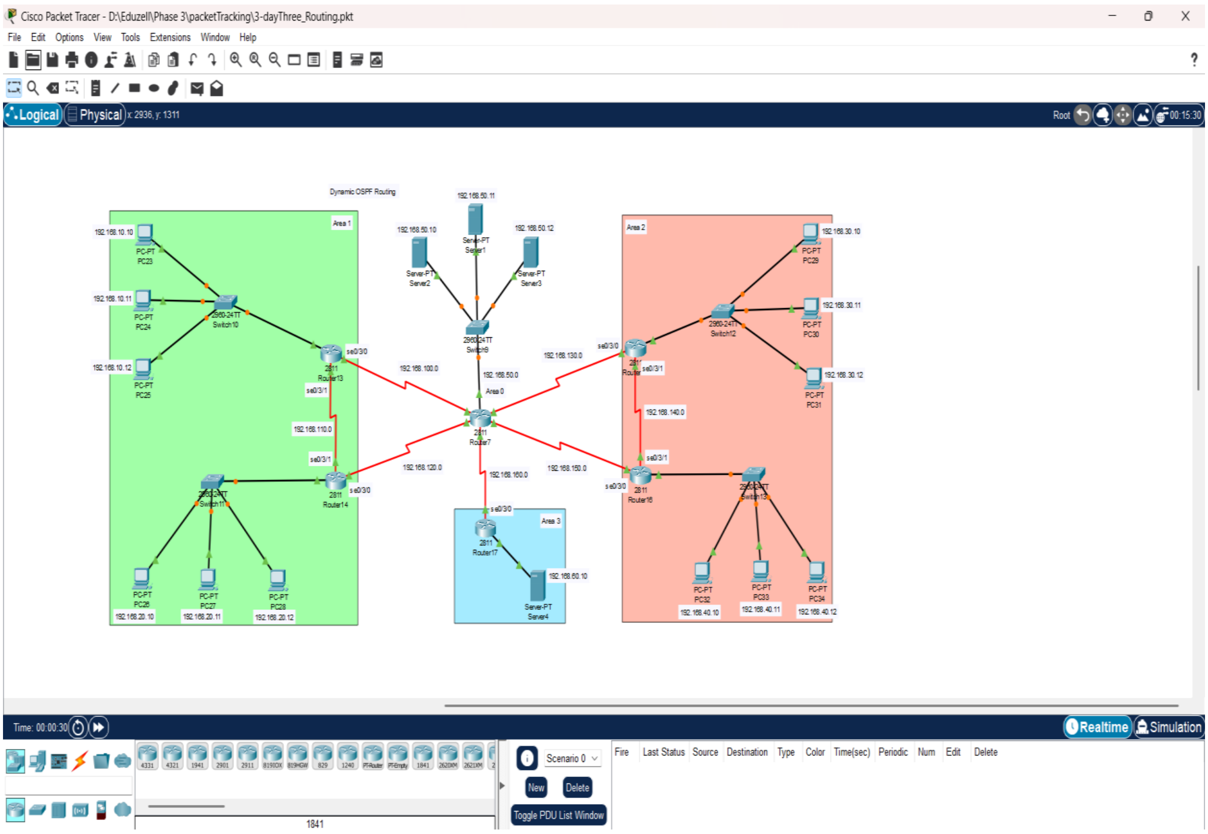
Lab 3.1 — RIP v2 (Routing Information Protocol)
Objective: Configure RIP v2 dynamic routing across a multi-router topology
Devices Used: 4x Cisco Routers, 3x Switches, 7x PCs
Key Commands: router rip, version 2, network, no auto-summary, show ip route, show ip rip database

- Built a multi-router topology with three separate network segments
- Configured RIP version 2 on all routers and advertised connected networks
- Disabled auto-summary to support classless routing across different subnets
- Monitored RIP route propagation using show ip route and observed RIP-learned routes (R)
- Tested full end-to-end reachability across all three segments
- Observed RIP convergence time and understood its limitations compared to link-state protocols



Lab 3.2 — OSPF (Open Shortest Path First)
Objective: Configure OSPF dynamic routing and compare behaviour against RIP
Devices Used: 12x Cisco Routers, 5x Switches, 6x PCs, 5x Servers
Key Commands: router ospf 1, network [ip] [wildcard] area 0, show ip ospf neighbor, show ip route

- Configured OSPF process on all routers within a single area (Area 0 — backbone)
- Advertised all connected networks using the network command with wildcard masks
- Verified OSPF neighbour adjacencies using show ip ospf neighbor
- Analysed the OSPF routing table and observed OSPF-learned routes (O)
- Compared OSPF convergence speed against RIP — observed faster, more scalable behaviour
- Tested full network reachability after OSPF convergence



Module 4: Access Control Lists (ACL) & Firewall Rules

This module covers the implementation of ACLs to control and filter network traffic at the router level. From a penetration testing perspective, understanding ACL logic is essential — both for bypassing poorly configured rules and for advising clients on correct implementation.

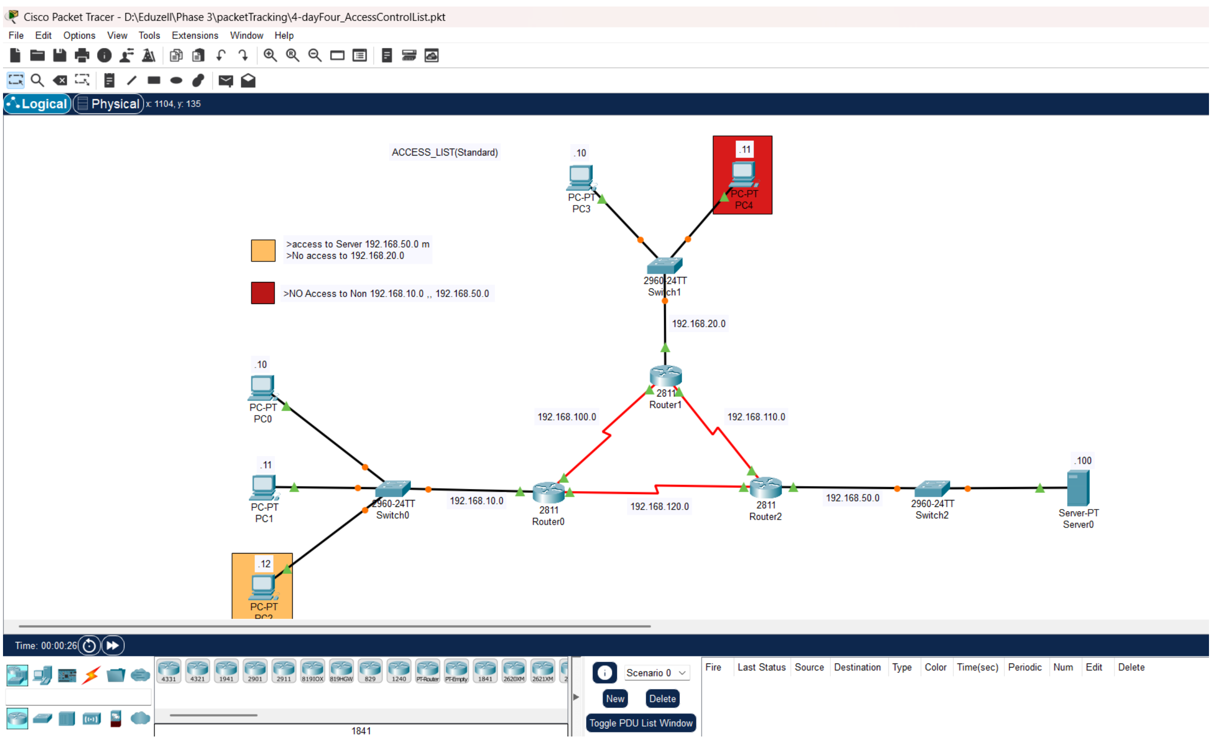
Lab 4.1 — Standard ACL

Objective: Configure a Standard ACL to permit or deny traffic based on source IP

Devices Used: 3x Switches, 5x PCs, 1x Server

Key Commands: access-list [1-99] permit/deny, ip access-group [in/out], show access-lists

- Created a Standard ACL to deny traffic from a specific source IP address/subnet
- Applied the ACL to a router interface in the outbound direction
- Tested the rule by attempting to ping the restricted destination from the blocked host — confirmed traffic was denied
- Verified that other hosts not covered by the deny rule could still communicate normally
- Used show access-lists to confirm the ACL was being hit (match count incrementing)



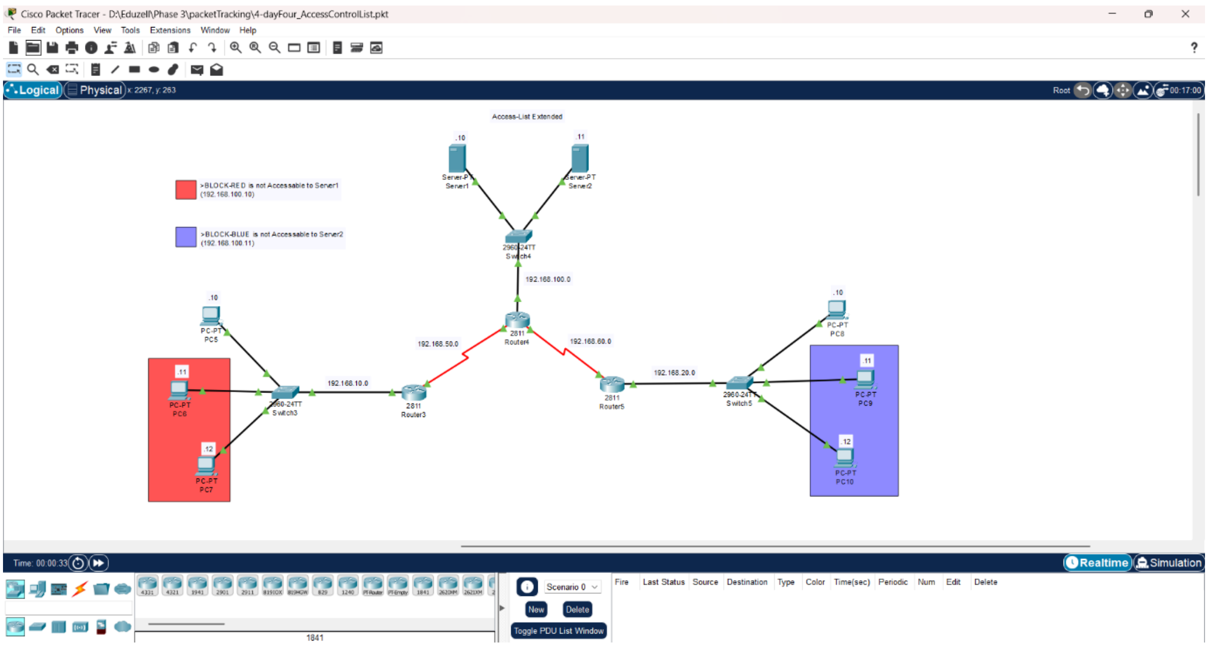
Lab 4.2 — Extended ACL

Objective: Configure an Extended ACL to filter traffic based on source IP, destination IP, and protocol/port

Devices Used: 2x Cisco Routers, 2x Switches, 4x PCs, 1x Server

Key Commands: access-list [100-199] permit/deny [protocol] [src] [dst] [port], show access-lists

- Created an Extended ACL to block HTTP traffic (TCP port 80) from a specific host to a web server
- Permitted all other traffic from the same host to pass through unrestricted
- Applied the ACL as close to the source as possible (inbound on the source-side interface)
- Tested by attempting HTTP access from the blocked host — confirmed connection refused
- Confirmed ICMP (ping) from the same host still worked — validating selective protocol filtering
- Used show access-lists to verify match counts for both permit and deny entries



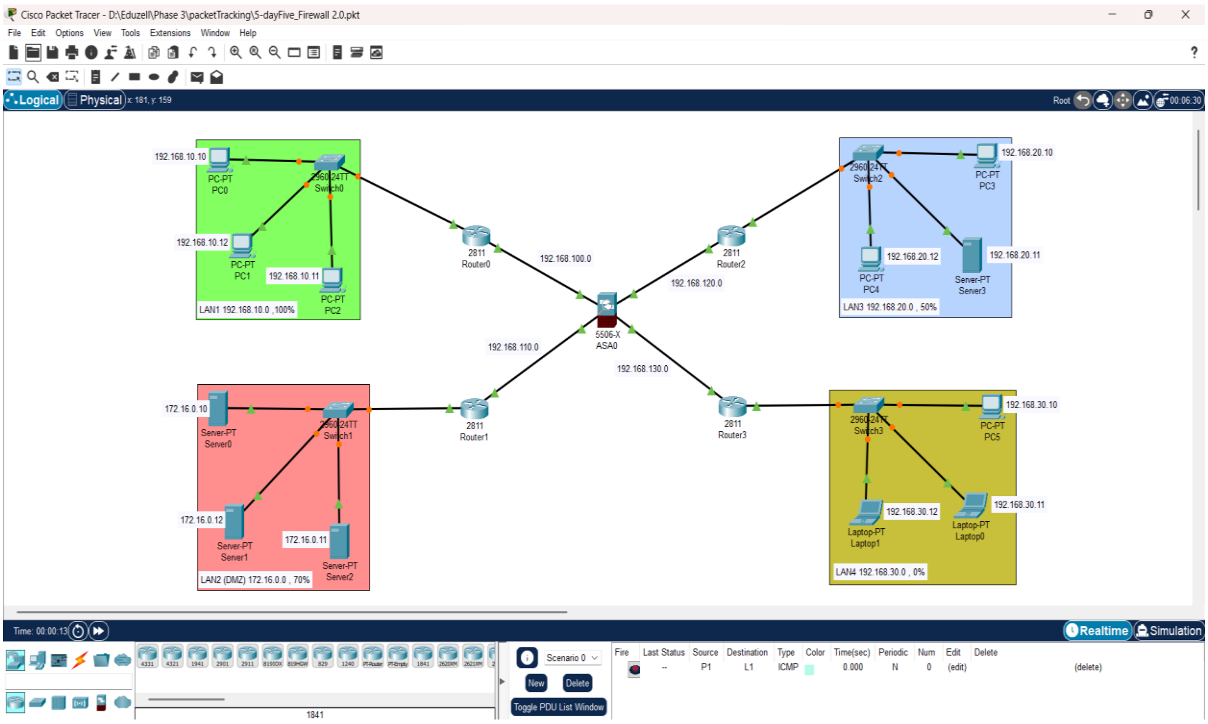
Lab 4.3 — Named ACL & Firewall Simulation

Objective: Implement a Named ACL to simulate a firewall policy protecting a server segment

Devices Used: 4x Cisco Routers, 4x Switches, 7x PCs, 4x Server

- Created a Named Extended ACL to simulate a basic firewall policy for a server segment
- Defined rules using remark comments for documentation — professional ACL writing practice
- Permitted only specific traffic types (e.g., HTTP, HTTPS, ICMP) to reach the server
- Denied all other traffic with an implicit final deny statement
- Applied the ACL to the router interface facing the server segment

Tested all permitted and denied traffic types to validate the firewall policy worked as intended



Result & Outcomes

Total Labs Completed	9 of 9
Modules Covered	Basic LAN Setup, VLAN, Routing Protocols, ACL & Firewall
Protocols Practised	OSPF, RIP v2, Standard ACL, Extended ACL, Named ACL
Connectivity Tests	All pass — end-to-end reachability verified across all labs
ACL Filtering Verified	Traffic permit/deny behaviour confirmed via match counts and connectivity tests
Overall Outcome	All objectives achieved successfully

Skills Demonstrated

- LAN design and IP addressing — foundation for all network-based assessments
- VLAN segmentation — understanding of broadcast domain isolation and VLAN hopping risks
- Inter-VLAN routing — essential for understanding enterprise network segmentation
- OSPF and RIP configuration — supports network enumeration and route analysis during pentests

  ACL implementation — understanding traffic filtering logic from both defensive and offensive perspectives

Conclusion

The completion of 13 structured Cisco Packet Tracer labs represents a deliberate and disciplined investment in the networking fundamentals that underpin all network-based penetration testing. From basic LAN connectivity to VLAN segmentation, dynamic routing, and access control implementation.

Understanding how networks are designed, segmented, and secured is not just a defensive skill — it is the foundation of effective offensive security.